

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (CE) Examination****May 2017****CE219.01 Object Oriented Programming using Java****Date: 31.05.2017, Wednesday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I**Q – 1 (A) State following statements are True or False. Also Justify your answer. [04]**

1. System should be divided in subsystem such that less interface are required.
2. In layered architecture lower layer work as a client.
3. Composition supports encapsulation.
4. A model is an abstraction of something for the purpose of understanding it before building it.

(B) Is UML a programming language? Pen down the names all the UML Diagrams. Classify each of the UML diagrams into structural group and behavioral group. [03]**(C) What is subsystem? How to identify the subsystem? How subsystems are interacting with each other? List out the relationships between subsystems. [02]****(D) List purpose of modeling. [02]****Q - 2 Answer the following questions. (Any two) [12]**

- (A) What do you mean by Layer? Explain with example. Also write brief note on two forms of layered architecture.
- (B) Which symbol is used to represent a class in class diagram? Also explain the parts of it. List the types of relations in class diagram, also mention their symbols and explain with example.
- (C) What is object oriented? List four concepts of OO explain them briefly.

Q – 3 Answer the following questions.**(A) Draw state diagram of telephone. [07]****(B) Name two kind of control flow in software system. Explain them in detail. [05]****OR****Q – 3 Answer the following questions.****(A) Draw the class diagram for the following system. [04]**

A library management system have three types of books: Journals, magazines and study books. Librarian can issue books and create transaction for the same. There are two types of members of library: student and faculty. Members can request any kind of book.

(B) Draw use case diagram of ATM system. [04]**(C) Draw activity diagram of Process-Order functionality of Box-office [04]**

SECTION – II**Q – 4 Answer the following short questions.**

- (A) Which data type in Java has the exact range from -2^{31} to $2^{31}-1$, inclusive? [01]
- (B) List out different ways to create thread. [01]
- (C) Define following terms. [02]
1. Bytecode
 2. Final variable
- (D) What is constructor? Can constructor be overloaded? Program has a class “Box” and it does not have constructor. In main method an object is created as “Box mybox = new Box()”. What happen in this situation? [02]
- (E) What is the output of following code? [02]

```
class GenMethDemo
{
    static <T extends Comparable<T>, V extends T> boolean isIn(T x, V[] y)
    {
        for(int i=0; i < y.length; i++)
            if(x.equals(y[i])) return true;
        return false;
    }
    public static void main(String args[])
    {
        Integer nums[] = { 1, 2, 3, 4, 5 };
        if(isIn(2, nums))
            System.out.println("2 is in nums");
        if(!isIn(7, nums))
            System.out.println("7 is not in nums");
        String str[] = { "one", "two", "three", "four", "five" };
        if(isIn("two", str))
            System.out.println("two is in str");
        if(!isIn("seven", str))
            System.out.println("seven is not in str");
    }
}
```

- (F) What is multitasking? Also write note on process based and thread based multitasking. [03]

Q - 5 Answer the following questions. (Any three)**[12]**

- (A) Write a program to transfer data from one file to another file so that if the destination file does not exist, it is created.
- (B) List similarities and differences between Array List and Vector.
- (C) Write a short note on lambda expression, lambda operator and block lambda.
- (D) What is package? Pen down access protections of class members for same package.
- (E) Diffentiate method overloading and method overriding with appropriate example.

Q - 6 Answer the following questions.

- (A) Write a Java program that will grade multiple choice test. Assume there are eight students and ten questions. Take answer key as input into one-dimensional array. Take answers of students as input and save into two-dimensional array. Compare each student's answer with the key, count number of correct answers and display result of each student on screen. [05]
- (B) Draw exception hierarchy. What happen when exceptions are not handled? Explain how to handle exception using try-catch, multiple catch and nested try in Java. [05]
- (C) What is the importance of InputStreamReader and OutputStreamWriter class in java? [02]

OR**Q - 6 Answer the following questions.**

- (A) Write a Java program to find summation of 1 to N numbers using threads. Take value of N and number of thread to be created from user. [05]
- (B) Write a Java program that illustrate the use of interface reference. Interface Luminious Object has two method lightOn() and lightOff(). There is one class Solid extended by 2 classes Cube and Cone. There is one class LuminiousCone extends Cone and implements Luminoius Interface. LumminuiousCube extends Cube and implements Luminious Interface. Create an object of LuminiousCone and LuminousCube and use the concept of interface reference to invoke the methods of interface. [05]
- (C) Write output of following program. [02]

```
public class Deer
{
    public Deer()
    {
        System.out.print("Deer");
    }
    public Deer(int age)
    {
        System.out.print("DeerAge");
    }
    private boolean hasHorns()
    {
        return false;
    }
    public static void main(String[] args)
    {
        Deer deer = new Reindeer(5);
        System.out.println(", " + deer.hasHorns());
    }
}
class Reindeer extends Deer
{
    public Reindeer(int age)
    {
        System.out.print("Reindeer");
    }
    public boolean hasHorns()
    {
        return true;
    }
}
```
